

## **QA Lab Brasil**

# **SQL, Database & Data Quality**

Data validation, SQL, relational/NoSQL databases, ETL/ELT and pipeline quality.

## **Databases for QA**

QA uses databases to validate persistence, integrations, business rules and consistency between UI, API and storage. Query safely and avoid unintended data changes.

## **Essential SQL**

SELECT reads data; WHERE filters; ORDER BY sorts; GROUP BY aggregates; JOIN combines tables. Before UPDATE or DELETE, confirm environment, permissions and impact.

## **JOIN in practice**

INNER JOIN returns matching rows. LEFT JOIN keeps all rows from the left and adds matches from the right. They are useful for validating orders, customers and relationships.

## **Transactions and ACID**

Atomicity, Consistency, Isolation and Durability help explain guarantees in transactional databases. Concurrent scenarios and rollback deserve attention in critical operations.

## **Data Quality**

Data quality includes dimensions such as completeness, accuracy, consistency, uniqueness and freshness. A system may work technically while still delivering wrong data.

## **ETL, ELT, Data Lake and Warehouse**

Pipelines move and transform data between sources and destinations. QA can validate counts, schema, transformations, duplicates, reconciliation and freshness.

## **NoSQL**

NoSQL databases use models different from relational databases, such as documents and key-value. Testing should consider consistency, flexible schema and query patterns.

## **Test-data security**

Avoid real personal data when it is not necessary. Prefer synthetic data, masking and anonymization according to applicable policies and laws.

## **QA tip**

Compare data at different points in the flow; a correct UI value does not prove the entire integration is correct.